G.3 Gemma-7B

In this section, we present the results for the Gemma-7B-Instruct model. As shown in figure 22, the

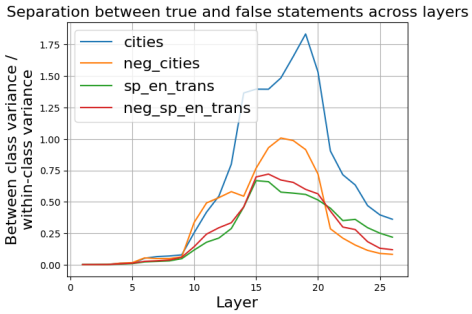
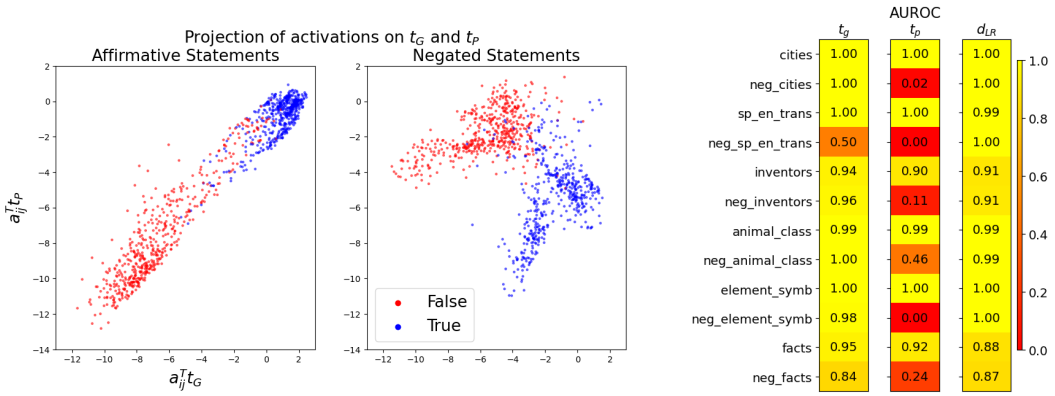


Figure 22: Gemma-7B: Ratio between the between-class variance and within-class variance of activations corresponding to true and false statements, across residual stream layers.

largest separation between true and false statements occurs in layer 16. Therefore, we use activations from layer 16 for the subsequent analysis of the Gemma-7B model. As can be seen in Figure 23, much higher classification accuracies would be possible by not only using $\mathbf{t}_G$ for classification but also $\mathbf{t}_P$.



(a) (b)

Figure 23: Gemma-7B: Left (a): Activations $\mathbf{a}_{ij}$ projected onto $\mathbf{t}_G$ and $\mathbf{t}_P$. Right (b): Separation of true and false statements along different truth directions as measured by the AUROC, averaged over 10 training runs.